Kanwar Pannu

Digital Design Engineer

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Summary

Computer Engineering graduate specializing in digital design, computer architecture, and hardware acceleration for AI. Experienced in RTL development and verification across CPUs, memory systems, and parametrizable AI accelerators, with hands-on experience spanning simulation, synthesis, FPGA implementation, and preliminary ASIC physical design. Through research and independent projects, I have explored novel AI architectures and memory systems while designing a 32-bit RISC-V CPU core from scratch and deploying a CNN accelerator on FPGA. I am passionate about designing the next generation of computing architectures that enable scalable, energy-efficient AI infrastructure and can contribute across the hardware stack from architectural design and RTL to verification, synthesis, and physical implementation.

Education

York University 2021–2026
BEng Computer Engineering Toronto, Canada

Experience

York University 2023–2025
Neuromorphics Researcher Toronto, Canada

- Designed and implemented spiking neural network (SNN) algorithms in Python/PyTorch to explore biologically inspired learning for low-power AI hardware.
- Collaborated with a multi-disciplinary team of researchers, preparing reports and internal documentation that translated latest neuroscience research into actionable engineering designs

Technical Skills

Languages: Python(Pandas, NumPy, ScikitLearn, cocotb), C, C++
Hardware Languages and Tools: Verilog, System Verilog, Verilator, ModelSim, Yosys, Quartus, LibreLane, OpenRoadflow, LTSpice, Icarus Verilog
Other: Makefile, Docker, Git, Linux, Jupyter, Bash

Projects

Accuracy-Aware Parametrizable TPU
https://github.com/Kanwarpartap-Pannu/TensorGrid

- Designed parameterizable MAC adders and arithmetic datapaths for a TPU supporting INT8/UINT8, INT4/UINT4, and INT2/UINT2 inference datatypes. Verified a UART module using Cocotb/Python, developing automated simulation models and functional testbenches.
- Synthesized RTL using Yosys and performed preliminary ASIC physical implementation/routing using the OpenLane/OpenROAD flow targeting GF180MCU. Gained hands-on experience across the RTL → verification → synthesis → physical design ASIC implementation flow.

ZCLLC - Last Level Cache for Embedded Systems
https://github.com/Kanwarpartap-Pannu/ZeroCost-LLC

- Developing a SystemVerilog implementation of a last-level cache (LLC) based on the research paper "High Performance and Predictable Shared Last-Level Cache for Safety-Critical Systems" (ACM TECS, 2024).
- Parametrizable set associative cache with LRU replacement policy. Write-Back and Write-Allocate Cache. Provides predictable timing for embedded systems without increasing worst case latency by removing back-invalidations.

5-Stage Fully Pipelined RISC-V CPU Core
https://github.com/Kanwarpartap-Pannu/RISCV-CORE

- Designed and implemented a 32-bit in-order 5-stage RISC-V CPU in SystemVerilog, including IF, ID, EX, MEM, and WB pipeline stages.

- Implemented hazard detection, data forwarding, and branch control logic to ensure correct execution under pipeline stalls and flushes.

FPGA Hardware Implementation of Convolutional Neural Network

- Designed and implemented a convolutional Spiking Neural Network in Verilog, including datapath, control logic, and memory interfaces, deployed on the DE10-Lite FPGA. Built automated verification flow using Icarus Verilog with Python-driven testbenches and Makefiles to validate functional correctness before synthesis.
- Integrated UART-based host FPGA communication for real-time image input and classification output, end-to-end hardware inference on MNIST.

ECG Monitoring and AI Anomaly Detection System

https://www.youtube.com/watch?v=GBLd_bZqjc4

- Designed and simulated a custom MOSFET-based analog front-end for ECG signal acquisition in LTspice, interfaced with Arduino for digital sampling. Validated real-world signal integrity using oscilloscope measurements and iterative hardware debugging.
- Built an end-to-end system streaming live ECG data to a Python-based ML inference pipeline and real-time web dashboard for anomaly detection.

Capstone Project - Tree Motion Energy Harvesting & IoT Monitoring System

- Designed embedded firmware on ESP32 to acquire and process voltage, temperature, humidity, and soil moisture data from multiple sensors.
- Built a real-time monitoring pipeline to analyze harvested energy performance and environmental conditions during live operation.